

Saurabh Kumar

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SUMMARY

Computer Science student with expertise in full-stack web development and artificial intelligence. Proficient in building production-ready applications using MERN stack, TensorFlow, and cloud technologies. Active open-source contributor with demonstrated success in developing scalable solutions achieving 95% model accuracy and handling 10,000+ daily API requests.

EDUCATION

Dr. APJ Abdul Kalam Technical University

Bachelor of Technology in Computer Science and Engineering; CGPA: 7.39/10.0

Lucknow, India

Aug. 2023 – May 2027

Uttar Pradesh Board (UPMSP)

Secondary School Certificate (Class X); Percentage: 87%

Uttar Pradesh, India

May 2020

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, C++, HTML5, CSS3, SQL

Frontend Technologies: React.js, Vite, Tailwind CSS, Responsive Design, Single Page Applications (SPA)

Backend Technologies: Node.js, Express.js, RESTful APIs, JWT Authentication, Session Management

Databases: MongoDB, MySQL, MongoDB Atlas, Query Optimization

Machine Learning: TensorFlow, PyTorch, Google Gemini AI, Deep Learning, LLM, Computer Vision

DevOps and Tools: Git, GitHub, Docker, Vercel, Render, Postman, Jupyter Notebook, npm, Nodemon

Core Competencies: Data Structures, Algorithms, System Design, OOP, Cloud Deployment

PROJECTS

StudyMate – AI-Powered Study Platform | *React, Node.js, MongoDB, Gemini AI*

Dec. 2024

- Engineered full-stack educational platform serving 20+ active users with AI-powered study assistance using MERN stack and Google Gemini API.
- Implemented intelligent chatbot for personalized learning with automated document summarization supporting PDF and DOCX formats.
- Developed comprehensive analytics dashboard featuring streak tracking and progress visualization using Recharts library.
- Integrated secure OTP-based email verification system using Resend API with bcrypt password encryption and session management.
- Established DDoS protection with rate limiting (3-100 requests per endpoint) and security headers using Helmet.js middleware.
- Deployed scalable architecture on Vercel (frontend) and Render (backend) with MongoDB Atlas for reliable data persistence.
- Links:** GitHub | Live Demo

Brain Tumor Detection System | *React, Node.js, Python, TensorFlow, Keras*

Nov. 2024

- Developed medical AI diagnostic system for brain tumor detection from MRI scans using Convolutional Neural Networks (CNN).
- Trained deep learning model achieving 95%+ accuracy on medical imaging dataset using TensorFlow and Keras frameworks.
- Created RESTful API with Express.js enabling real-time MRI image analysis with confidence score predictions and result visualization.
- Implemented secure hospital authentication system with bcrypt password hashing and role-based session access control.
- Integrated Python ML model with Node.js backend using child processes for seamless prediction workflow and error handling.
- Designed responsive React interface with drag-and-drop MRI upload, real-time preview, and detailed analysis result display.
- Links:** GitHub

HavenTo – Accommodation Booking Platform <i>React, Node.js, MongoDB, Express</i>	Oct. 2024
• Built full-stack booking platform handling 500+ registered users with property listings and comprehensive reservation management. • Developed RESTful APIs processing 10,000+ daily requests with robust error handling, input validation, and response optimization. • Implemented user authentication, favorites system, and booking functionality with secure JWT-based session management. • Integrated Cloudinary API for optimized image storage, upload, and delivery with automatic compression and caching. • Designed responsive user interface with React Router enabling seamless navigation and enhanced user experience across devices. • Deployed production application on Vercel (frontend) and Render (backend) with MongoDB Atlas for scalable data storage. • Links: GitHub Live Demo	

ACHIEVEMENTS

Contributed to 5+ open-source projects including React.js ecosystem and TensorFlow.js with merged pull requests.
Developed and deployed 3 production-ready full-stack applications on cloud platforms (Vercel, Render, AWS).
Achieved 95%+ accuracy in medical AI model for brain tumor detection from MRI scans using deep learning techniques.
Built scalable RESTful APIs handling 10,000+ daily requests with optimized performance and comprehensive error handling.
Solved 100+ algorithmic problems on LeetCode and GeeksforGeeks demonstrating strong problem-solving abilities.
Participated in college hackathons and technical competitions, securing recognition for innovative technology solutions.

CERTIFICATIONS

Front End Web Development <i>Online Course</i>	Nov. 2025
• Link: View Certificate	
Large Language Model <i>Online Course</i>	Feb. 2026
• Link: View Certificate	